

A Guide to Modern FPGAs

FPGAs are growing fast in technology and density. The latest advances have generally helped the design engineer to handle complex designs in FPGAs with better accuracy in lesser time. Let us see how it is done



Microsemi IGLOO2 FPGA

ANAGHA P.

Design architects are now more comfortable selecting a field-programmable gate array (FPGA) as the central piece on the board. Given the quantum of vendor and community resources available at their disposal, FPGAs makes it easier for them to complete their design within stipulated timelines and budget.

FPGAs are now capable of serving as suitable alternatives to application-specific integrated circuit (ASIC) or application specific standard product (ASSP) implementations in many systems—so much so that some vendors are marketing their FPGAs as ASIC-like.

Neeraj Varma, director of sales at Xilinx explained in an interview, “The whole belief is that the FPGAs in UltraScale generation have a lot many more features that are ASIC-like. The

key here is the ASIC-like class architecture. There are several changes we made in the architectures that are like ASICs. ASICs are always good in power and performance.

“FPGAs, traditionally, have an overhead, but we are trying to bridge this gap, and we did quite well in 28nm. But in 20nm, we are going to do even better.”

Earlier, FPGAs had few global clocks. As the density grew, skew problem also increased. Vendors, such as Xilinx, added more clocking regions, resulting in more clocking resources placed at different areas of the chip. This is how exactly it is done in ASICs to solve skew problem, and this is the reason why it is called an ASIC-class clocking.

Here, the software used is also an ASIC-strength design suite. The design suite has many built-in features where you are not allowed to make any mis-

takes, similar to how it is done for ASIC design tools. The tool itself takes care of so many things, saving your valuable time in repeating the design process.

Denser FPGAs

One of the points I came across in a white paper from Altera was that, because of shrinking of design cycles and design teams, cost-conscious designers were trying to move away from multi-chip solutions that were too costly and lacked flexibility. They were also moving away from single-chip solutions that could not meet power or performance objectives, or ASIC systems on chips (SoCs) which are too slow to get to market.

ARM-based SoCs combine a hard ARM processor, memory controllers and peripherals with customisable FPGA fabric in a single SoC. Initially, FPGAs were not dense enough and did not have hard cores on them.

Now, even for a single family of product, the user has quite a few options to select from, such as ARM-based SoCs that are user-customisable. This helps reduce system power and board size while increasing system perfor-



Virtex C2 AE-5 TXT FPGA from Xilinx